



Microwave radar for moisture detection

Researchers at Oak Ridge National Laboratory (ORNL) use microwave radar to detect moisture levels in walls without removing drywall or cladding, enabling early detection of mould growth. This technology is based on radar systems that emit microwave signals, which penetrate materials and



ORNL researchers set up instruments to direct microwave radar toward a sample of particle board, which represents the first inside layer, or sheathing, of a building's envelope. The test proved that this method can nondestructively detect moisture in the earliest stages of development (Credit: Carlos Jones/ORNL, US Dept. of Energy)

create reflections dependent on moisture content. Tested on wood sheathing samples, the radar system's readings were compared to conventional moisture meters, confirming its accuracy. The team is now working to commercialise the portable system, which could be used for inspecting walls, roofs, and foundations, helping homeowners and inspectors identify hidden moisture issues in buildings.

Electro-scattering antennas for advanced diagnostics

MIT researchers have developed a biosensing device featuring wireless, light-sensitive antennas that detect electrical signals between cells. This system, using organic electro-scattering antennas (OCEANs) made from PEDOT: PSS,



To improve biosensing techniques that can aid in diagnosis and treatment, MIT researchers developed tiny, wireless antennas that use light to detect minute electrical signals in liquid environments, which are shown in this rendering (Credits: Marta Airaghi and Benoit Desbieres, MIT)

offers high-resolution, efficient monitoring of cellular activity by measuring light intensity changes. Unlike traditional wired systems, it enables simultaneous data recording from thousands of points, enhancing the study of cellular communication in liquid environ-

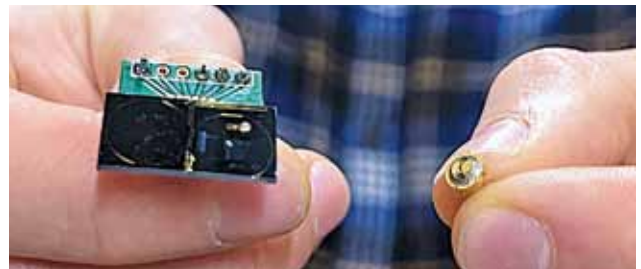
ments. With continuous recording capabilities for over 10 hours, OCEANs are particularly valuable for in vitro studies. The device is scalable, with sensors that detect faint signals (2.5mV), and shows promise for future integration with nanophotonic technologies.

Real-time visualisation of stretchable electronics

A POSTECH research team has developed a real-time technology to analyse and visualise the deformation of serpentine structures, vital components in stretchable electronics. Unlike previous methods, which required post-damage analysis, this system uses chiral liquid crystal elastomer (CLCE), a mechanochromic material that shifts colour during stretching, providing real-time deformation insights. This advancement, validated by finite element analysis, enables precise evaluation of structural behaviour without complex nanofabrication. The technology supports the design of more efficient, flexible systems for stretchable electronics, including displays, sensors, semiconductors, electronic skin, smart clothing, and soft robotics, helping to accelerate commercialisation in these industries.

Rubidium stabilises laser for quantum technology

At UC Santa Barbara, researchers have developed a chip-scale, ultra-low-line width 780nm laser, which could revolutionise atomic and quantum applications. By integrating



Ultra-high-quality ring resonator (left), which can help turn the 'coarse' light from a commercially available Fabry-Perot laser diode (right) into a low linewidth laser (Credit: Sonia Fernandez)

a commercially available Fabry-Perot laser diode with advanced waveguides and resonators on a silicon nitride platform, this compact laser matches the performance of bulky lab lasers, achieving improved frequency noise and linewidth stability. The device is ideal for quantum systems, portable sensors, and space-based applications, such as gravitational mapping and atomic timekeeping. It offers a significant cost reduction using a \$50 diode and CMOS-compatible fabrication, making it scalable for widespread scientific research and technology deployment.